

# Co-firing copper in a cordierite-type glass-ceramic printed by multi-material vat photopolymerisation: a model-based process design

[First Author]<sup>a,\*</sup>, [Second Author]<sup>a</sup>

<sup>a</sup>[Department, Institution], [City], [Country]

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## Abstract

Multi-material vat photopolymerisation can print a glass-ceramic body and its copper conductors as one green part, but whether such a part can be fired is not known. Carbon must leave the glass before its pores close, copper must stay metallic, and the two materials must shrink compatibly, all in one temperature range. A verified model couples pyrolysis and steam gasification of char, viscous sintering and crystallisation of the glass, diffusional sintering of copper, and co-firing mechanics, analytically and on the printed voxel grid. Inputs come from the literature, and the glass was calibrated against the published IBM glass-ceramic/copper process, whose atmosphere and burnout temperature the model reproduces. Only steam, with hydrogen held at a fixed fraction of the Cu/Cu<sub>2</sub>O equilibrium, removes the char before the glass seals. Fine copper then densifies during the burnout, well before the glass, and the free-strain mismatch reaches 15.6%. A paste and programme optimised together use the retarding effect of char on copper to release both materials at once, cutting the mismatch to 3.7%, but this optimum is fragile. A design that tolerates a factor of two in gasification rate and 15 K in the glass transition holds the mismatch at 9–11% and the copper at 89% IACS in a 16 h cycle. In three dimensions the optimum reduces the dishing of a printed test vehicle from 622 to 150  $\mu\text{m}$ , and plastic flow of the copper keeps the cooling bow small. Gasification kinetics, glass viscosity, copper sintering viscosity and char yield should be measured first.

**Keywords:** Co-firing, Glass-ceramic, Copper metallisation, Vat photopolymerisation, Binder removal, Sintering mismatch

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## 1. Introduction

Packages for millimetre-wave front ends, antenna-in-package modules and wide-bandgap power devices now have to route conductors in three dimensions through a low-loss dielectric, and to do it close to the die [1, 2, 3, 4]. Co-fired ceramics answer part of this demand. Low-temperature co-fired ceramic (LTCC) and glass-ceramic substrates combine low dielectric loss, hermeticity and a thermal expansion close to that of silicon [5, 6]. They are, however, built from stacked tapes: conductors are printed in planes, interconnections run vertically through punched vias, and every change of geometry requires new tooling [7, 5]. The Heterogeneous Integration Roadmap lists additive manufacturing of ceramic packages, including lithography-based ceramic manufacturing, as one way past these limits [8].

The conductor of choice in fired packages has long been silver, because it can be co-fired in air. Silver migrates under bias and humidity and diffuses into the glass when fired in oxidising conditions [9, 10, 11]. Copper conducts almost as well, costs less and is far less prone to migration, but it must be fired in an atmosphere that keeps it metallic. In such an atmosphere the organic binder no longer burns off cleanly: part of it is left as char, and char trapped in a glass that has already sealed its pores degrades dielectric loss and can bloat the part [12]. IBM solved this for its glass-ceramic/copper multilayer substrates

with a steam-based co-sintering process that gave a dense glass-ceramic with high-conductivity copper [13, 6]. That solution was developed for laminated tapes and its operating window was found empirically.

Vat photopolymerisation of ceramic slurries, in particular lithography-based ceramic manufacturing (LCM), produces dense ceramics with features of a few tens of micrometres [14, 15, 16]. It has already been used for monolithic microwave filters and antenna substrates [17, 18, 19], and photocurable LTCC suspensions have been formulated for it [20]. In these parts the metallisation is still added afterwards, by plating or by printing on the fired ceramic [17, 21], or the conductors are jetted into ceramic layers by a separate route [22, 23, 24, 25]. Multi-material vat photopolymerisation removes that restriction. Printers such as the CeraFab Multi 2M30 switch between slurries within a layer [26, 27, 28]. Co-sintered ceramic–ceramic parts have been demonstrated [29], as has a zirconia part with titanium-oxide conductors co-fired in a reducing atmosphere [30]. Copper slurries for vat photopolymerisation have been developed separately [31, 32, 33, 34]. In principle, then, a glass-ceramic body and its copper wiring can be printed as one green part. We found no study of whether such a part can be fired.

Four requirements make that firing difficult, and they are coupled. First, the char from both slurries must be removed before the glass seals its pores, in an atmosphere that does not oxidise the copper. Second, the two materials must shrink at compatible times and by compatible amounts; otherwise the part cambers, delaminates or cracks around its conductors [35,

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\*Corresponding author.

Email address: [email address] ([First Author])

36, 37]. Third, the glass must densify before it crystallises, because crystals stop viscous flow. Fourth, a thermal-expansion mismatch of about  $14 \text{ ppm K}^{-1}$  between copper and a cordierite-type glass-ceramic must be accommodated on cooling. In tape-based LTCC the second requirement is met by trial, most often by adding ceramic powder to the metal paste [38] or by adjusting the heating rate [39]. Continuum models of co-sintering reproduce camber and mismatch stresses in ceramic and metal–ceramic bilayers [40, 41, 42, 43, 44, 45], but none includes the chemistry of binder removal, and none treats copper with a crystallising glass. The first and third requirements are therefore usually handled separately from the second, although in practice they compete for the same temperature range.

This paper sets out a model-based design of that firing for a copper conductor printed inside a cordierite-type glass-ceramic body. We extend a verified slab model of binder removal and sintering, originally developed for printed copper, to a crystallising glass-ceramic, and couple the two materials through the viscous analogue of laminate theory and a three-dimensional finite-element model built on the printed voxel grid. The model is verified against closed-form solutions and parameterised from published data, with every input tagged by its provenance. With it we (i) map the atmosphere and temperature window in which carbon can be removed from the glass before its pores close while copper stays metallic; (ii) show that fine copper and a fragile glass densify in different temperature ranges and with different curve shapes, and quantify the consequences for camber and conductor damage; (iii) design the copper paste and the firing programme together to bring the two into step, both at nominal inputs and robustly against their uncertainty, and follow the designs through a printed test vehicle in three dimensions; and (iv) identify the few material parameters that decide the outcome, and should therefore be measured first.

## 2. Model formulation

### 2.1. Structure of the model

The co-fired part is represented at two levels (Fig. 1b). At the first level, each material is followed through the firing programme by a one-dimensional slab model that resolves heat transfer, binder decomposition, char formation and removal, pore closure and densification through the thickness of its critical section. The glass-ceramic body and the copper conductor are simulated separately but share the furnace set point and the inlet gas, so both see the same thermal and chemical history. At the second level, the two densification histories are coupled mechanically: analytically, through the viscous analogue of laminate theory, and numerically, through a three-dimensional finite-element model in which both materials occupy the voxels of the printed part. Both levels use the same constitutive law for the porous body, so a free sample in the finite-element model reproduces the slab model exactly.

The copper slab model was developed and verified for single-material copper parts printed from a lithography-based slurry; its chemistry and its calibration against published copper data are summarised in Appendix A. The additions made here are the

crystallising glass-ceramic, the copper paste with an inert filler, the steam burnout atmosphere, and the co-firing mechanics.

### 2.2. Binder removal and residual carbon

The photocured binder is lumped into three pseudo-components  $i$  (a non-reactive solvent, a crosslinked network and a char-forming backbone), each decomposing by first-order kinetics,

$$\frac{dB_i}{dt} = -k_i(T) B_i, \quad k_i(T) = \frac{E_i \beta_{\text{ref}}}{RT_{p,i}^2} \exp\left[-\frac{E_i}{R} \left(\frac{1}{T} - \frac{1}{T_{p,i}}\right)\right], \quad (1)$$

where  $T_{p,i}$  is the temperature of the corresponding peak in a thermogravimetric trace at  $\beta_{\text{ref}} = 10 \text{ K min}^{-1}$ . Writing the rate constant in terms of the peak temperature keeps the two fitted quantities nearly independent and lets them be read directly from a single thermogravimetric curve. A fraction  $\chi$  of the network and backbone mass remains as char. In a steam-bearing atmosphere the char is gasified by  $\text{C} + \text{H}_2\text{O} \rightarrow \text{CO} + \text{H}_2$  at the rate

$$r_g = a(\varepsilon_o) (1 - f_{\text{cl}}) k_g(T) \frac{c}{M_C} \left(\frac{x_{\text{H}_2\text{O}}}{x_{\text{ref}}}\right)^{1/2} \times \frac{1 + K_H x_{\text{H}_2}^{\text{ref}}}{1 + K_H x_{\text{H}_2}} \left(1 - \frac{Q}{K_{\text{st}}(T)}\right)_+, \quad (2)$$

in which  $c$  is the char concentration,  $a(\varepsilon_o)$  the fraction of pore surface reachable through open porosity  $\varepsilon_o$ ,  $f_{\text{cl}}$  the closed-pore fraction,  $K_H$  an inhibition constant for hydrogen adsorption,  $Q = x_{\text{CO}} x_{\text{H}_2} / x_{\text{H}_2\text{O}}$  the reaction quotient and  $K_{\text{st}}$  the equilibrium constant from NIST-JANAF data [46]. Char in closed pores is not gasified: once the pores seal, whatever carbon remains stays in the part. The rate constant  $k_g$  is parameterised by a half-life at  $800^\circ\text{C}$  in the reference atmosphere and an activation energy; no measured value exists for char on either material, so both enter the sensitivity analysis. A Thiele-modulus estimate ( $k_g L^2 / D_{\text{eff}} \sim 10^{-4}$  for  $L = 2 \text{ mm}$ ) shows that gasification is kinetically controlled, so steam access is represented by the open-pore accessibility  $a(\varepsilon_o)$  rather than by a diffusion equation.

Pyrolysis gas raises the pore pressure. The overpressure is computed from the compressible Darcy equation with Kozeny–Carman and Knudsen contributions to the permeability and a polymer-permeation term that carries the flux before the pore network percolates; it is compared with the green or brown strength through a safety factor. The same slab tracks the through-thickness temperature difference and converts it into a thermal-stress index.

### 2.3. Densification

Both materials densify according to the Skorohod–Olevsky viscous sintering model [47]. For a porous body of porosity  $\theta$  and matrix shear viscosity  $\eta$ , the stress is

$$\sigma = 2\eta [\varphi(\theta) \dot{\varepsilon}' + \psi(\theta) \text{tr}(\dot{\varepsilon}) \mathbf{I}] + P_L(\theta) \mathbf{I}, \quad (3)$$

with  $\varphi = (1 - \theta)^2$ ,  $\psi = \frac{2}{3}(1 - \theta)^3 / \theta$  and sintering stress  $P_L = 3\gamma(1 - \theta)^2 / r$ , where  $\gamma$  is the surface energy and  $r$  half the Sauter

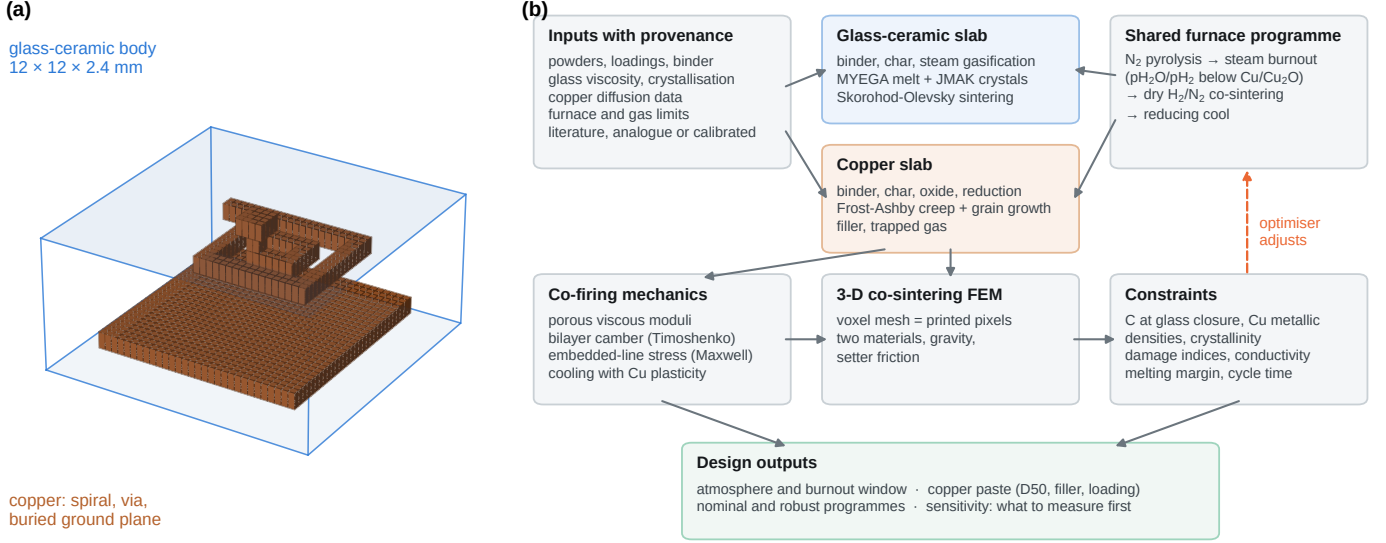


Figure 1: (a) Test vehicle: a 12 × 12 × 2.4 mm glass-ceramic body with a buried copper ground plane, a square spiral and a via to a top pad, printed as one part. (b) Structure of the model. Each material is followed by its own slab model under a shared furnace programme; the two histories are coupled mechanically, analytically and in three dimensions, and the programme and paste are searched against constraints.

mean particle diameter. A free body densifies at  $\text{tr}(\dot{\epsilon}) = -(P_L - \Delta p)/(2\eta\psi)$ , where  $\Delta p$  is the pressure of insoluble gas trapped in closed pores. The two materials differ only in  $\eta$ ,  $\gamma$  and  $r$ .

*Glass-ceramic.* The residual melt is Newtonian. Its viscosity follows the MYEGA equation [48],

$$\log_{10} \eta_g = \log_{10} \eta_\infty + (12 - \log_{10} \eta_\infty) \frac{T_g}{T} \times \exp \left[ \left( \frac{m}{12 - \log_{10} \eta_\infty} - 1 \right) \left( \frac{T_g}{T} - 1 \right) \right], \quad (4)$$

with  $\log_{10} \eta_\infty = -3$  (Pa s), glass transition  $T_g$  (defined at  $10^{12}$  Pa s) and fragility index  $m$ . Crystallisation during firing follows non-isothermal Johnson–Mehl–Avrami–Kolmogorov kinetics [49] written in terms of the extended-volume integral,

$$\dot{\xi} = k_c(T) = k_{c,p} \exp \left[ -\frac{E_c}{R} \left( \frac{1}{T} - \frac{1}{T_{p,c}} \right) \right], \quad X = 1 - \exp(-\xi^n), \quad (5)$$

which avoids the singularity of the rate form at  $X = 0$  for  $n > 1$ . The constant  $k_{c,p}$  is fixed numerically so that the simulated crystallisation exotherm at  $10 \text{ K min}^{-1}$  peaks exactly at the measured temperature  $T_{p,c}$ ; Kissinger’s closed form [50] is exact only for  $n = 1$  and would misplace the peak by about 1 K for  $n = 2$ . Crystals stiffen the melt as rigid inclusions,

$$\eta = \eta_g(T) \left( 1 - \frac{X}{X_m} \right)^{-[\eta]X_m} \left( 1 - \frac{f_c}{0.64} \right)^{-[\eta]0.64}, \quad (6)$$

the Krieger–Dougherty law [51] with intrinsic viscosity  $[\eta] = 2.5$  and maximum packing  $X_m$ . The second factor treats residual char particles, volume fraction  $f_c$ , the same way. Once  $X$  approaches  $X_m$  the body stops flowing, which is how crystallisation arrests densification.

*Copper.* The copper skeleton creeps by diffusion. Its matrix viscosity combines lattice and grain-boundary diffusion [52],

$$\eta_{Cu} = f_\eta \frac{k_B T G^2}{42 \Omega (D_v + \pi \delta D_b / G)} \left( 1 + \frac{C^2}{C_{inh}^2} \right) \left( 1 - \frac{f}{0.64} \right)^{-1.6}, \quad (7)$$

where  $G$  is the grain size,  $\Omega$  the atomic volume and  $f_\eta$  a calibration factor fitted to published copper densification data (Appendix A). The carbon term  $C$  (in ppm) describes the retarding effect of residual carbon on copper, and the last factor accounts for an inert ceramic filler of volume fraction  $f$  added to the conductor paste, a standard way to slow a metal paste down to the pace of its ceramic [38]. Grain growth is parabolic, driven by grain-boundary diffusion and pinned by porosity. The filler also lowers the conductivity; together with porosity,

$$\frac{\sigma}{\sigma_0} = (1 - \theta)^{3/2} \frac{1 - f}{1 + f/2}, \quad (8)$$

the second factor being the Maxwell Garnett result for insulating inclusions [53].

#### 2.4. Atmosphere and the metallic state of copper

Steam removes char but oxidises copper once the water-to-hydrogen ratio exceeds the equilibrium of  $\text{Cu}_2\text{O} + \text{H}_2 \rightleftharpoons 2\text{Cu} + \text{H}_2\text{O}$ ,

$$\frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}} < K_{\text{Cu}}(T) = \exp \left( -\frac{\Delta G^\circ(T)}{RT} \right). \quad (9)$$

During the burnout segment the inlet hydrogen is therefore tied to the steam content,  $x_{\text{H}_2} = x_{\text{H}_2\text{O}}/(s K_{\text{Cu}}(T_B))$ , with a safety factor  $s = 0.2$ . The copper stays metallic by construction, and the steam is as oxidising towards carbon as the metal allows. This is the principle of the steam co-sintering developed for glass-ceramic/copper substrates [13, 6]; here it is applied in

a model that says how long, at what temperature and in how much steam. Carbon gasification itself is thermodynamically unconstrained over the whole window (Section 4.2); its limits are kinetic.

## 2.5. Mechanics of co-firing

*Porous viscosities.* Eq. (3) describes an isotropic, linear viscous solid with shear viscosity  $G_v = \eta\varphi$  and bulk viscosity  $K_v = 2\eta\psi$ , hence uniaxial viscosity  $E_v = 9K_v G_v / (3K_v + G_v)$ , viscous Poisson ratio  $\nu_v = (3K_v - 2G_v) / [2(3K_v + G_v)]$  and biaxial viscosity  $B = E_v / (1 - \nu_v)$ . The free linear sintering rate is  $\dot{\epsilon}_f = -P_L / (6\eta\psi)$ . All three quantities follow from the slab histories of  $T$ ,  $\theta$ ,  $G$  or  $\xi$ , and carbon.

*Camber.* By the elastic–viscous correspondence principle, a bilayer of two linear viscous layers bends at a curvature rate given by Timoshenko’s bimetal solution [54], with elastic moduli replaced by biaxial viscosities and thermal strain by the difference in free sintering rates [55, 37, 40]:

$$\dot{\kappa} = \frac{6(1 + \lambda)^2 (\dot{\epsilon}_{f,2} - \dot{\epsilon}_{f,1})}{h[3(1 + \lambda)^2 + (1 + \lambda\mu)(\lambda^2 + 1/(\lambda\mu))]}, \quad (10)$$

with  $\lambda = h_1/h_2$ ,  $\mu = B_1/B_2$  and  $h = h_1 + h_2$ . Layer 1 is copper and layer 2 the glass-ceramic; positive curvature is concave towards the copper. Eq. (10) is integrated over the firing with the layer thicknesses updated by their own shrinkage. The reference stack is a 0.1 mm copper plane on 1 mm of glass-ceramic.

*Embedded conductors.* A conductor line buried in the body cannot curl; its misfit is taken up by stress. We represent line and surrounding body as two Maxwell elements in series, each with a porous elastic modulus and a biaxial viscosity:

$$\dot{\epsilon}_{f,GC} - \dot{\epsilon}_{f,Cu} = \dot{\sigma} \left( \frac{1}{E_{Cu}} + \frac{1}{E_{GC}} \right) + \sigma \left( \frac{1}{B_{Cu}} + \frac{1}{B_{GC}} \right). \quad (11)$$

A compliant line follows the body and carries  $B_{Cu}$  times the misfit rate; a stiff line forces the body to flow around it. Tension in the copper ( $\sigma > 0$ ) opens voids and breaks the line; compression in the copper means the body is shrinking onto a stiff inclusion and is itself in hoop tension, the classic origin of radial cracks around conductors [35, 36]. Two damage indices compare these stresses with the strength of the weaker partner at that moment,

$$\Pi_{Cu} = \frac{\max(\sigma, 0)}{\sigma_{s,Cu}(\rho_{Cu})}, \quad \Pi_{GC} = \frac{\max(-\sigma, 0)}{\sigma_{s,GC}(\rho_{GC})}, \quad (12)$$

where the strengths  $\sigma_s$  grow from the binder-carried green strength through the neck-controlled brown strength. The series coupling carries an Eshelby factor of order unity [56] that we do not resolve, so the indices are used comparatively. The same coupling is resolved exactly in three dimensions below.

*Three-dimensional co-sintering.* The printed part is meshed voxel by voxel with trilinear hexahedra, so the mesh is the printed pixel grid. Each element carries its material, porosity and microstructural state. The velocity field satisfies  $\nabla \cdot \sigma + \rho_b \mathbf{g} = \mathbf{0}$  with Eq. (3), gravity, and a setter plate beneath the part: zero

normal velocity and regularised Coulomb friction ( $\mu = 0.6$ ) iterated to convergence at each step. Volumetric terms use one-point quadrature, which avoids locking as  $\theta \rightarrow 0$ . The problem is assembled with scikit-fem [57] and solved by conjugate gradients preconditioned with smoothed-aggregation algebraic multigrid carrying the six rigid-body modes [58]. Porosity is advanced with the exact mass-conserving update  $1 - \theta^{n+1} = (1 - \theta^n) \exp(-\text{tr } \dot{\epsilon} \Delta t)$ . The temperature and carbon histories are taken from the two slab simulations. Below 0.3% porosity the constitutive law is evaluated at 0.3%, since the bulk viscosity otherwise grows without bound and the remaining densification no longer affects the shape. The simulation ends when the glass-ceramic reaches 98% density or 90% crystallinity, whichever comes first; the remaining linear shrinkage of the body is then a few tenths of a per cent.

*Cooling.* Below a stress-free temperature  $T_{sf}$ , where the residual glass stops relaxing, the thermal-expansion mismatch is carried elastically by the glass-ceramic and elastoplastically by the copper. The copper layer is treated as elastic–perfectly plastic with a temperature-dependent yield stress; its mean stress is capped at yield by a uniform plastic strain, and the curvature follows from Timoshenko’s elastic solution for the remaining elastic misfit.

## 2.6. Numerical solution, design search and sensitivity

The slab equations are integrated with a linearly implicit Rosenbrock method of orders 2(3) [59] on a geometrically graded mesh that is finest at the surface. A co-firing evaluation, two slab simulations and the mechanical post-processing, takes about 6 s. Designs were searched by differential evolution [60] over eleven variables: copper median particle size, filler fraction and solids loading; burnout temperature, time and steam fraction; temperature and duration of an intermediate matching hold; heating rate to the peak; peak temperature and hold. The objective was the largest free-strain mismatch between the two materials during firing,  $\max_t |\epsilon_{f,GC} - \epsilon_{f,Cu}|$ , which governs both camber and conductor stress. It was minimised subject to hard constraints and to a cycle-time budget of 18 h, the  $\epsilon$ -constraint form of the trade-off between cycle time and mismatch. The constraints were: carbon at glass pore closure below 100 ppm and in the copper below 200 ppm; relative densities of at least 0.97 for the glass-ceramic and 0.92 for the copper; a crystallised fraction of at least 0.8; a line damage index below 1; no trapped-gas pressure above 1.2 times the sintering stress; and a positive margin to the melting point of copper. Violations entered the objective as a penalty proportional to their size, so the search is steered back towards feasible designs. The population of 88 designs was evolved for 15 generations.

A second, robust search asked for a design that works across the uncertainty of the two inputs that decide the timing, the gasification rate and the glass transition. It evaluates each design at two corners of that uncertainty: slow gasification ( $t_{1/2} = 4$  h) with a low transition ( $T_g - 15$  K), where the glass seals before the char has gone, and fast gasification ( $t_{1/2} = 1$  h) with a high transition ( $T_g + 15$  K), where the copper is released before the glass is ready. The objective is the larger of the two peak mismatches; all constraints must hold at both corners, and the

copper must in addition reach 80% IACS. The cycle-time budget was relaxed to 20 h, because the first feasible designs found needed about 20 h; the final design needs 16 h, so the budget did not bind. The search was seeded with one such feasible design, and a population of 66 was evolved for eight generations. The nominal case lies between the corners and was checked on the final design.

The influence of uncertain inputs was screened with Morris elementary effects [61]: 16 trajectories through a four-level grid over fourteen factors, 240 co-firing evaluations in total, around the optimised design. Twelve factors are the uncertain inputs with the ranges of Table 1; the other two are manufacturing tolerances on the copper paste,  $-25$  to  $+33\%$  in particle size and  $\pm 0.05$  in filler fraction. For each output the mean absolute elementary effect  $\mu^*$  measures the overall influence of a factor and the standard deviation  $\sigma$  its non-linearity and interaction with other factors.

### 3. Parameters and calibration

#### 3.1. Provenance of the inputs

No measurements were made for this study. Every input therefore comes from one of five sources, and Table 1 states which one applies. A value is labelled L when it was reported for the glass-ceramic of the IBM process or for copper itself; A when it was taken from an analogous material, usually a commercial copper-compatible LTCC; C when it was calibrated here against published process observations (Section 3.2); D when it follows from other entries of the table; and E when it is an engineering estimate. Estimated and analogue values that matter enter the sensitivity analysis of Section 4.10, with the ranges given in the table.

The reference glass is the MgO-rich cordierite-type glass used by IBM for its copper-compatible substrates (glass #12: 52.5 SiO<sub>2</sub>, 22.0 MgO, 22.0 Al<sub>2</sub>O<sub>3</sub>, 1.5 P<sub>2</sub>O<sub>5</sub>, 0.5 B<sub>2</sub>O<sub>3</sub>, 1.5 ZrO<sub>2</sub>, wt%) [62, 63]. It was chosen because it is the only copper-compatible glass-ceramic for which the firing process, the atmosphere and the reasons behind the chosen temperatures have been published in detail. Its viscosity has not been published in a form we could read, so it was calibrated.

#### 3.2. Calibration of the glass against the IBM process

The patents that describe the IBM steam co-sintering process state, for glass #12, why the burnout hold sits where it does. The hold is placed at  $785 \pm 10$  °C, between the annealing and softening points of the glass. At 750 °C carbon removal would take “prohibitively excessive amounts of time”; at 830 °C the glass pores close and trap water and binder residue [62]. A later patent from the same group reports that 33–50% of the total fired shrinkage occurs during the burnout step [78], and a third places the onset of crystallisation at about 900 °C, after the burnout hold and before the crystallisation hold at about 960 °C [79]. We simulated the full worked schedule of Ref. [62] (N<sub>2</sub> to 200 °C, H<sub>2</sub>/H<sub>2</sub>O to 450 °C, 2.9 K min<sup>-1</sup> to the hold, 6 h at H<sub>2</sub>/H<sub>2</sub>O = 10<sup>-4</sup>, 0.5 h in N<sub>2</sub>, 2.1 K min<sup>-1</sup> to 960 °C, 2 h) and used these

statements as targets. The patent specifies the H<sub>2</sub>/H<sub>2</sub>O ratio but not the dilution; we assumed 50% steam in nitrogen.

Two of them fix the viscosity. Requiring that a 6 h hold at 780 °C produce 40% of the fired shrinkage with the pores still open, and that the pores close within 2 h at 830 °C, gives  $\log_{10} \eta = 10.62$  at 780 °C and 9.37 at 830 °C. The MYEGA curve through these two points, with  $\log_{10} \eta_{\infty} = -3$ , has  $T_g = 733.6$  °C and  $m = 32.4$ . The remaining observations were not used in the fit and serve as checks. Between 800 and 860 °C the calibrated viscosity falls by one decade per 43 K. Giess et al. found that the sintering of an IBM cordierite-type glass over the same range followed the viscosity of the glass, which fell by about one decade per 40 K [65]. The annealing point, taken as 10<sup>12</sup> Pa s, lies at 734 °C and the dilatometric softening point (10<sup>6.6</sup> Pa s) at 980 °C, which brackets the 785 °C hold as the patent requires. Stoichiometric cordierite glass is one to four decades more viscous over 780–960 °C [80]. Glass #12 contains more MgO and less Al<sub>2</sub>O<sub>3</sub> than cordierite, together with B<sub>2</sub>O<sub>3</sub> and P<sub>2</sub>O<sub>5</sub>, so a lower viscosity is plausible, but this comparison is not a test of the fit.

The crystallisation kinetics were treated in the same way. With the activation energy measured on an IBM-type glass ( $E_c = 303$  kJ mol<sup>-1</sup> [67]) and  $n = 2$ , placing the onset of crystallisation (5% crystallised) at 900 °C in the worked schedule requires an exotherm peak at 1040 °C at 10 K min<sup>-1</sup>. The simulated exotherm then spans 978–1066 °C, a width of about 90 K, which agrees with the 80–100 °C window that the IBM patent associates with the crystallisation exotherm of its glasses [63]. Differential thermal analysis of other MgO–Al<sub>2</sub>O<sub>3</sub>–SiO<sub>2</sub> glasses places the  $\alpha$ -cordierite peak at 1006–1032 °C [64, 69].

Table 2 summarises the result. The 785 °C hold removes the char and leaves 47% of the shrinkage to the burnout step, inside the 33–50% reported for the process, while the pores stay open. At 830 °C the body densifies completely during the hold and seals its pores. These two rows reproduce the calibration targets and are therefore not independent evidence. The 750 °C row is: it depends on the gasification kinetics, which were not fitted, and it leaves four orders of magnitude more carbon than the 785 °C hold. The model does not reproduce the absolute time scale of the patent’s statement. With the estimated gasification rate, 33 ppm of carbon remain after 6 h at 750 °C, which may or may not have been acceptable for IBM; no carbon specification was published. For all three holds the final body is 99% dense and fully crystallised, and its free linear shrinkage is 23.1%.

## 4. Results

#### 4.1. Verification

The model was checked against closed-form solutions before it was used for design (Appendix B). Isothermal densification of the glass-ceramic follows the Skorohod–Olevsky solution to within  $1.3 \times 10^{-4}$  in porosity for 4 to 16 cells (Fig. 2a), and the three-dimensional curvature rate of a free copper/glass-ceramic plate converges to the viscous Timoshenko value at second order, with an error of 0.05% at 0.15 mm voxels (Fig. 2b). The analytical camber of Eq. (10) and the finite-element model therefore

Table 1: Model inputs. Basis: L, literature value for this material; A, analogue material; C, calibrated in Section 3.2; D, derived; E, estimate. Ranges in the last column are those used in the Morris screening.

| Quantity                                                    | Symbol                         | Value                                  | Basis | Source, remark or screening range                                                                                                                                                      |
|-------------------------------------------------------------|--------------------------------|----------------------------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Glass-ceramic body</i>                                   |                                |                                        |       |                                                                                                                                                                                        |
| Median particle size                                        | $d_{50}$                       | $3.0 \mu\text{m}$                      | A     | IBM glass powders $2\text{--}7 \mu\text{m}$ [63]; $2\text{--}7 \mu\text{m}$                                                                                                            |
| Size span $(d_{90} - d_{10})/d_{50}$                        | –                              | 1.5                                    | E     |                                                                                                                                                                                        |
| Solids loading of the slurry                                | $\phi_{\text{GC}}$             | 0.45                                   | E     | 0.40–0.50                                                                                                                                                                              |
| Density of the dense body                                   | $\rho_s$                       | $2600 \text{ kg m}^{-3}$               | A     | dense $\alpha$ -cordierite glass-ceramic, $2620 \text{ kg m}^{-3}$ [64]                                                                                                                |
| Surface energy                                              | $\gamma$                       | $0.36 \text{ J m}^{-2}$                | L     | calculated for a cordierite-type glass [65]; 0.24–0.36, since water vapour lowers the surface tension of silicate glass [66]                                                           |
| Glass transition ( $10^{12} \text{ Pa s}$ )                 | $T_g$                          | $734^\circ\text{C}$                    | C     | $714\text{--}754^\circ\text{C}$                                                                                                                                                        |
| Fragility index                                             | $m$                            | 32.4                                   | C     | 28–40                                                                                                                                                                                  |
| High-temperature viscosity limit                            | $\log_{10} \eta_\infty$        | –3                                     | L     | [48]                                                                                                                                                                                   |
| Crystallisation activation energy                           | $E_c$                          | $303 \text{ kJ mol}^{-1}$              | L     | IBM-type high-cordierite glass [67]; $330\text{--}440 \text{ kJ mol}^{-1}$ in other $\text{MgO--Al}_2\text{O}_3\text{--SiO}_2$ glasses [68, 69]; $250\text{--}470 \text{ kJ mol}^{-1}$ |
| Avrami exponent                                             | $n$                            | 2                                      | E     | $1.5\text{--}3.7$ ; $2.8\text{--}3.7$ in a related $\text{MgO--Al}_2\text{O}_3\text{--SiO}_2$ glass [69]                                                                               |
| Exotherm peak at $10 \text{ K min}^{-1}$                    | $T_{\text{p,c}}$               | $1040^\circ\text{C}$                   | C     | $1000\text{--}1080^\circ\text{C}$                                                                                                                                                      |
| Maximum crystal packing                                     | $X_m$                          | 0.64                                   | E     | random close packing                                                                                                                                                                   |
| Density at pore closure                                     | $\rho_{\text{cl}}$             | 0.92                                   | E     |                                                                                                                                                                                        |
| Thermal conductivity, dense                                 | $k$                            | $2.8 \text{ W m}^{-1} \text{ K}^{-1}$  | A     | copper co-fired LTCC GL570 [70]                                                                                                                                                        |
| Young’s modulus, dense                                      | $E_{\text{GC}}$                | 128 GPa                                | A     | GL570 [70]                                                                                                                                                                             |
| Poisson’s ratio                                             | $\nu_{\text{GC}}$              | 0.25                                   | E     |                                                                                                                                                                                        |
| Thermal expansion coefficient                               | $\alpha_{\text{GC}}$           | $3.0 \text{ ppm K}^{-1}$               | L     | production glass-ceramic substrate [71]                                                                                                                                                |
| Flexural strength                                           | $\sigma_{s,\text{GC}}$         | 210 MPa                                | L     | glass #12 fired in $\text{H}_2/\text{H}_2\text{O}$ [62]                                                                                                                                |
| Stress-free temperature                                     | $T_{\text{sf}}$                | $690^\circ\text{C}$                    | D     | strain point ( $10^{13.5} \text{ Pa s}$ ) of the calibrated glass                                                                                                                      |
| <i>Copper conductor</i>                                     |                                |                                        |       |                                                                                                                                                                                        |
| Median particle size                                        | $d_{50}$                       | $1\text{--}20 \mu\text{m}$             | –     | design variable; baseline $3 \mu\text{m}$                                                                                                                                              |
| Size span                                                   | –                              | 1.2                                    | E     |                                                                                                                                                                                        |
| Solids loading                                              | $\phi_{\text{Cu}}$             | 0.40–0.60                              | –     | design variable; baseline 0.50                                                                                                                                                         |
| Inert filler fraction                                       | $f$                            | 0–0.40                                 | –     | design variable; baseline 0                                                                                                                                                            |
| Viscosity factor                                            | $f_\eta$                       | 0.28                                   | C     | Appendix A, from [31]; 0.1–1.0                                                                                                                                                         |
| Lattice and boundary diffusion                              | $D_v, \delta D_b$              | –                                      | L     | [52]                                                                                                                                                                                   |
| Carbon inhibition scale                                     | $C_{\text{inh}}$               | 200 ppm                                | E     |                                                                                                                                                                                        |
| Young’s modulus                                             | $E_{\text{Cu}}$                | 117 GPa                                | L     | oxygen-free copper [72]                                                                                                                                                                |
| Poisson’s ratio                                             | $\nu_{\text{Cu}}$              | 0.34                                   | E     |                                                                                                                                                                                        |
| Thermal expansion coefficient                               | $\alpha_{\text{Cu}}(T)$        | $16.5\text{--}23.7 \text{ ppm K}^{-1}$ | L     | instantaneous values, 293–1100 K [73]                                                                                                                                                  |
| Yield stress                                                | $\sigma_y(T)$                  | 69 MPa at $20^\circ\text{C}$           | L     | annealed temper [72]; softening $1 - T^{*1.09}$ [74, 75]                                                                                                                               |
| <i>Binder, char and atmosphere</i>                          |                                |                                        |       |                                                                                                                                                                                        |
| Char yield of network and backbone                          | $\chi$                         | 0.05                                   | E     | 0.02–0.10                                                                                                                                                                              |
| Gasification half-life, $800^\circ\text{C}$ , reference gas | $t_{1/2}$                      | 2 h                                    | E     | 0.5–8 h                                                                                                                                                                                |
| Gasification activation energy                              | $E_g$                          | $200 \text{ kJ mol}^{-1}$              | A     | order of steam gasification of carbons [76]; $160\text{--}240 \text{ kJ mol}^{-1}$                                                                                                     |
| Hydrogen inhibition constant                                | $K_H$                          | 25                                     | E     | form after [77]                                                                                                                                                                        |
| Cu/Cu <sub>2</sub> O and C/H <sub>2</sub> O equilibria      | $K_{\text{Cu}}, K_{\text{st}}$ | –                                      | L     | NIST-JANAF [46]                                                                                                                                                                        |
| Safety factor on the Cu/Cu <sub>2</sub> O limit             | $s$                            | 0.2                                    | E     |                                                                                                                                                                                        |
| <i>Geometry</i>                                             |                                |                                        |       |                                                                                                                                                                                        |
| Slab half-thickness                                         | –                              | $2.0 / 0.15 \text{ mm}$                | E     | glass-ceramic / copper                                                                                                                                                                 |
| Reference bilayer                                           | $h_{\text{Cu}}, h_{\text{GC}}$ | 0.1, 1.0 mm                            | E     | 20 mm span                                                                                                                                                                             |

Table 2: The IBM worked schedule [62] simulated at three hold temperatures. Shrinkage is the fraction of the total fired shrinkage reached by the end of the 6 h hold; carbon is the residual carbon in the glass-ceramic at the end of the hold.

| Hold temperature            | $750^\circ\text{C}$ | $785^\circ\text{C}$ | $830^\circ\text{C}$ |
|-----------------------------|---------------------|---------------------|---------------------|
| Shrinkage during hold       | 7%                  | 47%                 | 100%                |
| Pores at end of hold        | open                | open                | closed              |
| Carbon at end of hold (ppm) | 33                  | 0.006               | 0.07                |
| Final relative density      | 0.990               | 0.990               | 0.990               |
| Final crystallised fraction | 1.00                | 1.00                | 1.00                |
| Statement in [62]           | too slow            | chosen              | pores close         |

describe the same mechanics, and the analytical form can be used inside the design search.

#### 4.2. Atmosphere window for burnout

Fig. 3 places the burnout atmosphere in the plane of temperature and  $p_{\text{H}_2\text{O}}/p_{\text{H}_2}$ . Copper remains metallic below the orange line, which falls from  $10^{6.2}$  at  $500^\circ\text{C}$  to  $10^{4.8}$  at  $780^\circ\text{C}$  and  $10^{4.1}$  at  $1000^\circ\text{C}$ . Carbon gasification by steam is thermodynamically possible above the dashed line, which lies eight or more decades lower. The two constraints therefore leave a wide band of permissible atmospheres, and the choice within it is kinetic. At  $780^\circ\text{C}$ , 30% steam with hydrogen at 20% of the copper limit ( $p_{\text{H}_2\text{O}}/p_{\text{H}_2} = 1.2 \times 10^4$ ) gives a char half-life of 0.43 h. Forming gas (4%  $\text{H}_2$ ) humidified through a bubbler at  $20^\circ\text{C}$ , a common choice for copper, contains 2.3% water; its half-life at the same temperature is 3.1 h, seven times longer, because the steam partial pressure is low and hydrogen inhibits the reaction.

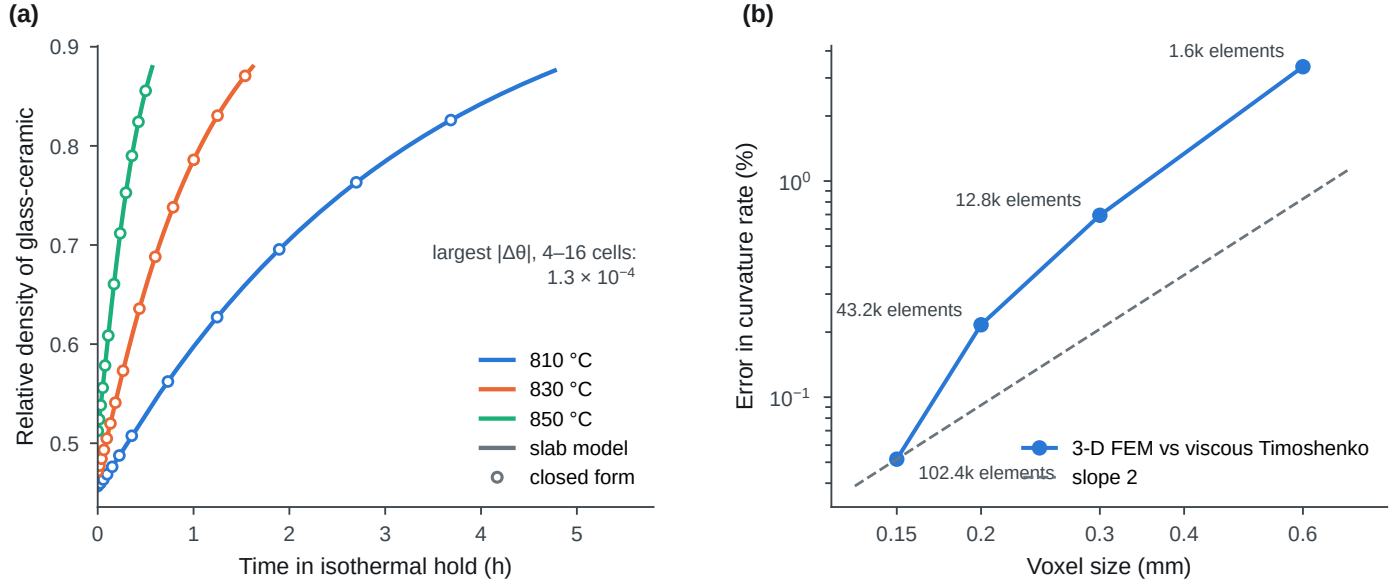


Figure 2: Verification. (a) Isothermal densification of the glass-ceramic slab at 810, 830 and 850 °C (lines, 8 cells) against the closed-form solution integrated along the simulated part temperature (circles); crystallisation and char were switched off. (b) Error of the three-dimensional curvature rate of a free 12 × 12 mm plate (0.6 mm copper on 1.8 mm glass-ceramic, 900 °C) against the viscous Timoshenko solution, Eq. (10), as the voxel size is refined.

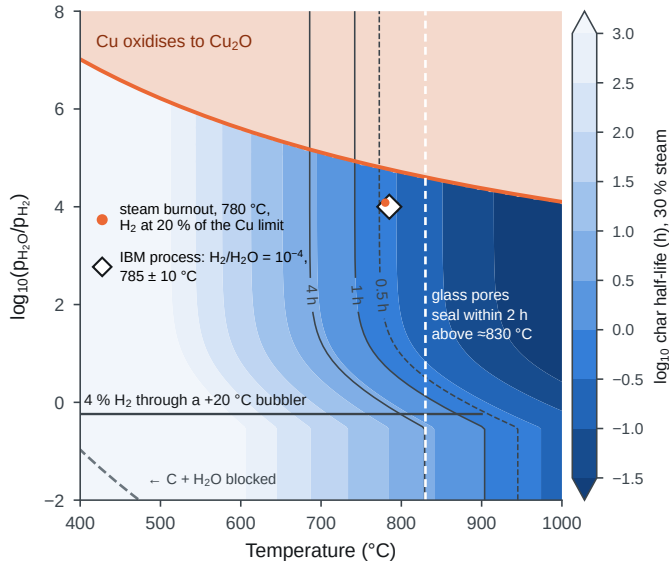


Figure 3: Burnout atmosphere window. Colour: half-life of char in 30% steam at the given  $p_{H_2O}/p_{H_2}$  (reference kinetics of Table 1). Copper oxidises above the orange line; carbon cannot be gasified to the left of the dashed line ( $p_{CO} = 10^{-4}$  atm). Circle: steam burnout used in this work. Diamond: operating point of the IBM process [62]. The white line marks the temperature above which a 2 h hold seals the pores of the calibrated glass.

The operating point of the IBM process,  $H_2/H_2O = 10^{-4}$  at  $785 \pm 10$  °C [62], lies at 17% of the copper limit and almost coincides with the steam point in Fig. 3. The rule used here, hydrogen tied to steam so that the water-to-hydrogen ratio sits at a fixed fraction of the copper limit, therefore reproduces the atmosphere that was found empirically for tape-cast substrates. It also says how the ratio should change if the burnout temperature changes: by a factor of about 2.5 per 100 K, following  $K_{Cu}(T)$ .

#### 4.3. Removing carbon before the glass seals

The char has to leave the glass before its pores close, and the glass carries far more of it than the copper does. After pyrolysis in nitrogen the glass-ceramic contains 22 500 ppm carbon relative to its solids, the copper 5 300 ppm. The difference follows from the densities: both slurries hold a similar volume of binder, but a unit volume of glass-ceramic solids weighs a third as much as the same volume of copper. Carbon on the glass is also the more harmful, since whatever is sealed in reacts with the water released during crystallisation and bloats the part [62].

Fig. 4 maps the carbon left in the glass-ceramic at the moment its pores close, as a function of burnout temperature and hold time. In steam (Fig. 4a) the 100 ppm contour runs from 16 h at 710 °C to 1.1 h at 830 °C, and the window closes abruptly at 850 °C, where the glass seals during the hold. The IBM hold of 6 h at 785 °C falls in the region with less than 10 ppm. In humidified forming gas (Fig. 4b) only the longest hold, 16 h at 790 °C, brings the carbon down to 350 ppm; every other combination leaves more than 1000 ppm. Residual carbon also slows the densification of the glass. Acting as rigid inclusions, the 0.3–2% of carbon retained in forming gas lowers the final density from 0.99 to 0.96–0.99. Steam is therefore not an option for this material pair but a requirement.

#### 4.4. Two densification ranges

Fig. 5 compares the free sintering of the two materials after binder removal at  $5 \text{ K min}^{-1}$  in dry hydrogen. The glass-ceramic shrinks by 23% between 850 and 933 °C (5 and 95% of its shrinkage) and then stops, because it crystallises: 93% of the glass has crystallised at 1060 °C. Fine copper densifies much earlier, and over a much wider range. A 3  $\mu\text{m}$  powder shrinks between 454 and 702 °C; a 1  $\mu\text{m}$  powder between 323 and 463 °C. Only for powders of 12  $\mu\text{m}$  or coarser does the midpoint

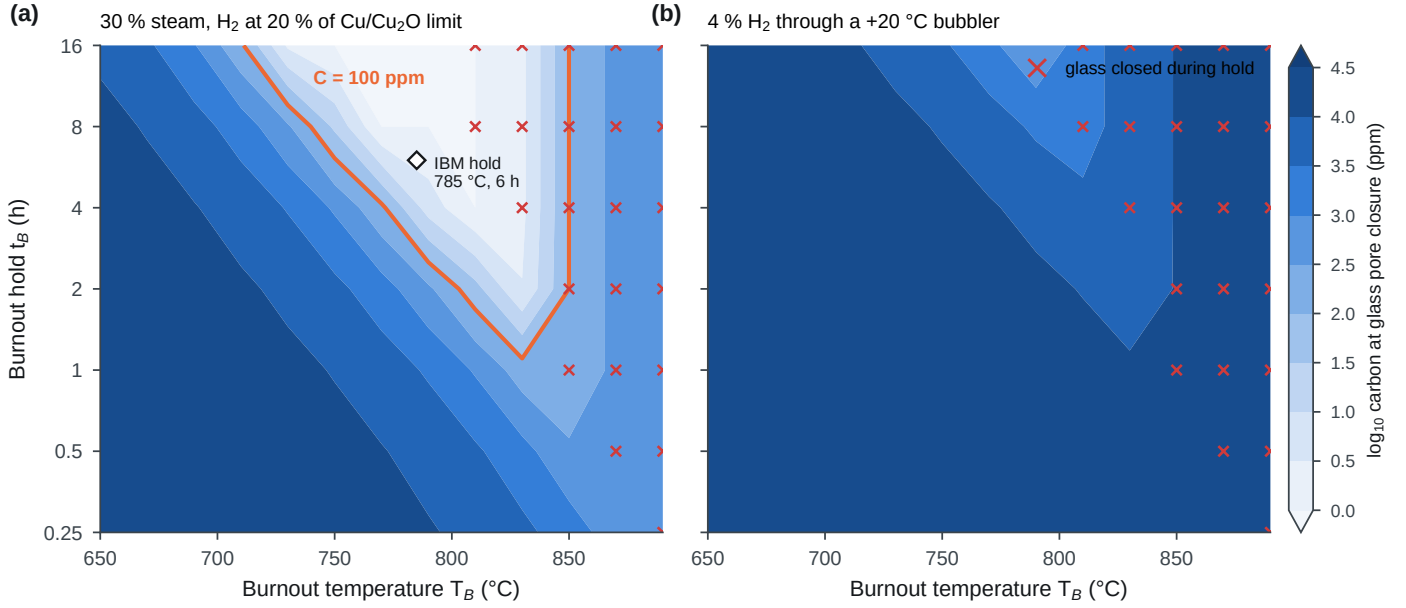


Figure 4: Carbon remaining in the glass-ceramic when its pores close, for burnout holds of temperature  $T_B$  and duration  $t_B$  within the baseline programme. (a) 30% steam with hydrogen at 20% of the  $\text{Cu}/\text{Cu}_2\text{O}$  limit; (b) 4%  $\text{H}_2$  humidified at 20 °C. Orange: 100 ppm contour. Crosses: the glass sealed during the hold. Diamond: the IBM burnout hold [62].

of densification fall inside the range of the glass, and these powders do not reach high density before 1060 °C: 83% for 12  $\mu\text{m}$  and 60% for 20  $\mu\text{m}$ .

The two curves also differ in shape. The glass densifies over a range of about 80 K, as a Newtonian melt whose viscosity falls steeply with temperature. Copper densifies over 140–350 K, because diffusional creep has a lower activation energy and grain growth slows it as it proceeds. Adding an inert filler to the paste delays and stretches the copper curve (Fig. 5b): 40% filler shifts the midpoint of 3  $\mu\text{m}$  copper from 573 to 657 °C and widens the 5–95% range from 248 to 367 K. Neither particle size nor filler, on its own, gives copper a shrinkage curve that is as late and as steep as that of the glass. The matching has to come partly from the furnace programme.

#### 4.5. Co-firing with an unmatched paste

The baseline design combines a 3  $\mu\text{m}$  copper paste without filler with a programme that follows the IBM pattern: pyrolysis in nitrogen to 460 °C, a 4 h burnout at 780 °C in 30% steam, and sintering at 960 °C for 1 h in dry 4%  $\text{H}_2$  (Fig. 6a). Chemically, the design works. The glass-ceramic retains 33 ppm of carbon when its pores close and reaches 99.1% density and full crystallinity; the copper is 97.6% dense, retains 81 ppm of carbon and conducts at 96% IACS.

Mechanically, it fails. As soon as steam begins to remove the char, the copper starts to sinter, while the glass is still rigid. At the end of the burnout hold the logarithmic shrinkage of the copper is 21% and that of the glass 7%; the free-strain mismatch peaks at 15.6%, and a free bilayer of 0.1 mm copper on 1 mm glass-ceramic would reach a curvature of 58  $\text{m}^{-1}$ , concave towards the copper. The damage index of an embedded line reaches  $\Pi_{\text{Cu}} = 1.6$  at 680 °C, during the ramp to the burnout

temperature, when the line is pulled in tension by the rigid body around it while its own necks are still weak. The glass then catches up. Its shrinkage overtakes that of the copper at 889 °C, the bilayer bends the other way and reaches  $-259 \text{ m}^{-1}$  (a radius of 3.9 mm) at 930 °C, and the final shrinkages differ by 4%: 26.3% for the glass-ceramic against 22.3% for the copper, both as logarithmic strains. For this material pair the difficulty lies less in the final shrinkage than in the order in which the two materials shrink.

#### 4.6. Copper paste design maps

Fig. 7 shows how far the materials alone can repair the baseline, with the programme left unchanged. The upper row varies the copper paste. Coarser powder and filler both delay the copper, and the damage index of an embedded line falls below 1 for particle sizes of 4.5  $\mu\text{m}$  and above (Fig. 7b). The peak mismatch falls more slowly. Its lowest values, about 9.5%, are reached at 6  $\mu\text{m}$  with 30% filler and at 9  $\mu\text{m}$  without filler; coarser pastes are released too late and the mismatch rises again, now with the copper behind the glass. Every step towards a later copper costs density or conductivity (Fig. 7c,d). Filler lowers the conductivity directly, by 14 points of IACS for every 10 vol%, while coarse powder lowers it through porosity: 18  $\mu\text{m}$  copper reaches only 76% density. Among the pastes that meet the density limit, 9  $\mu\text{m}$  copper without filler gives the best compromise: 9.5% peak mismatch, a damage index of 0.38 and 91% IACS. None of them reduces the final camber, which stays between  $-190$  and  $-260 \text{ m}^{-1}$  for every paste that meets the density limit. The final camber is set by the difference in final shrinkage, which the paste barely changes.

The lower row varies the glass, with the baseline paste. The glass transition decides whether carbon is trapped. Below about

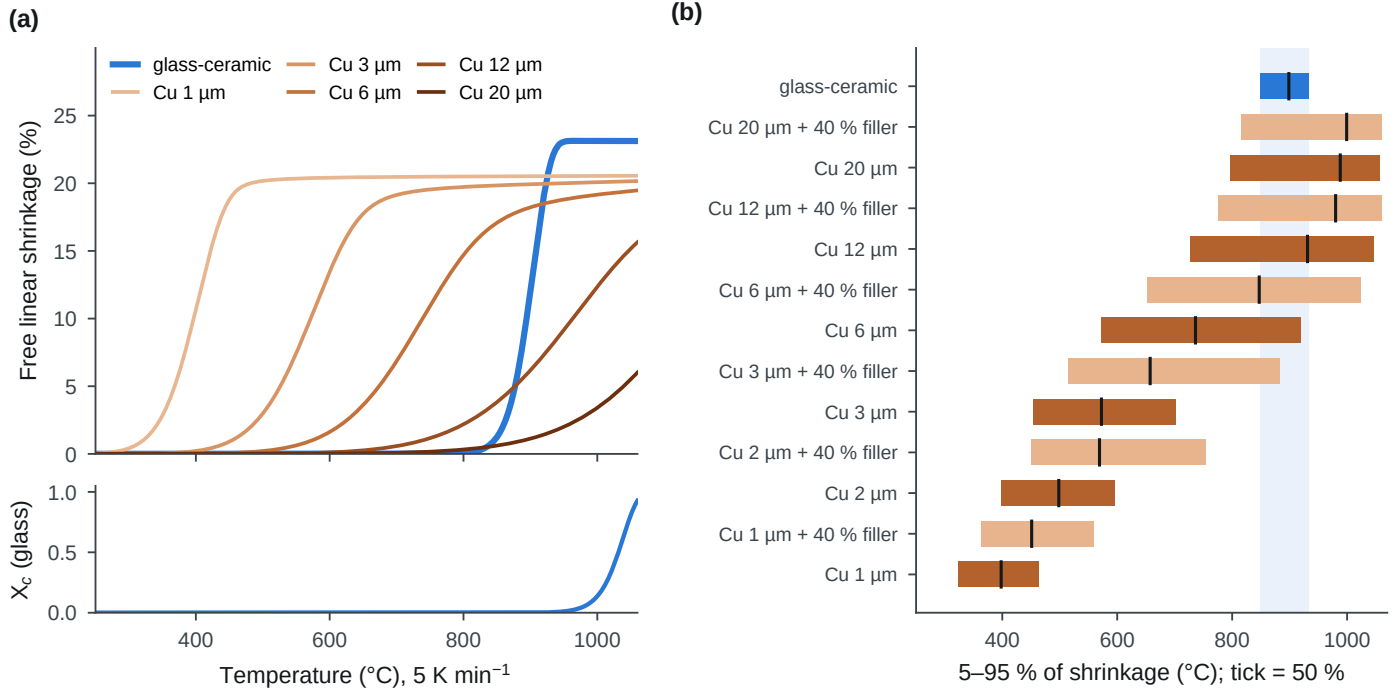


Figure 5: Free sintering of the two materials at 5 K min<sup>-1</sup> in dry 4% H<sub>2</sub> after binder removal. (a) Linear shrinkage of the glass-ceramic and of copper of different median particle size, and crystallised fraction of the glass. (b) Temperature range between 5 and 95% of the final shrinkage, with the midpoint marked, for copper with and without 40 vol% inert filler; the shaded band is the range of the glass-ceramic. Bars that reach 1060 °C had not finished densifying.

695 °C the glass seals during the 780 °C burnout and retains about 3000 ppm (Fig. 7e). Above that, the carbon stays below 100 ppm, but the higher the transition, the later the glass densifies, and the crystallisation peak has to rise with it if the glass is to reach full density before it crystallises (Fig. 7f). The crystallisation peak must not be too high either: above about 1080 °C the glass is less than 80% crystalline after 1 h at 960 °C (Fig. 7g). The most interesting region is the narrow band around  $T_g = 700$  °C (Fig. 7h). There the glass densifies during the burnout hold, together with the copper, and the peak mismatch falls to 6.5% while the carbon at closure stays near 50 ppm. A glass chosen to match the copper would thus do more than any paste, but only within a window of about 20 K in  $T_g$ , and with little margin on carbon at its lower edge.

#### 4.7. A matched paste and programme

Neither the paste nor the glass alone brings the peak mismatch below about 6%, so the paste and the programme were searched together (Section 2.6). The optimum found within an 18 h cycle is listed in Table 3 and its history is shown in Fig. 6b. It uses a 3.9 μm copper powder with 24 vol% filler. The burnout is short and hot: 1.3 h at 840 °C in 81% steam, just below the temperature at which the glass would seal during the hold (Fig. 4a). A 1.7 h hold at 916 °C follows, and then 1.2 h at 1018 °C. The peak mismatch falls from 15.6 to 3.7%, the damage index of an embedded line from 1.6 to 0.83, and the final camber of the reference bilayer from -254 to -76 m<sup>-1</sup>. The glass-ceramic is 99.0% dense and fully crystallised, with 0.2 ppm of carbon at pore closure.

The optimiser found a mechanism rather than a compromise. The copper cannot sinter while it carries char (Section 4.5), so the burnout releases it. Placing the burnout just below the coalescence of the glass, and making it fast with a high steam fraction, releases the copper at the moment the glass begins to flow. The two materials then densify together (Fig. 6b), and the filler slows the copper just enough to follow the glass rather than lead it. The hold at 916 °C lets the glass finish densifying before it crystallises, and the peak at 1018 °C finishes the copper.

The optimum has two weaknesses. The first is conductivity: 24% filler limits the copper to 64% IACS. The second is robustness. The coincidence on which the optimum rests is sharp, and Table 3 shows what happens when the two least certain inputs are varied within plausible bounds. If the char gasifies half as fast as assumed, the carbon at closure rises from 0.2 to 76 ppm and the peak mismatch from 3.7 to 11.1%; at a quarter of the assumed rate the carbon limit is exceeded by an order of magnitude. A glass transition 15 K higher or lower than calibrated raises the peak mismatch to about 10%. A factor of two in the viscosity of copper matters less (5.8–6.2%). A design that relies on the timing of char removal is therefore only as good as the gasification kinetics and the viscosity of the glass, and neither has been measured for this material pair.

The robust search (Section 2.6) therefore asked for a design that meets every constraint at two corners of this uncertainty and, in addition, keeps the copper above 80% IACS. The result, listed in the last column of Table 3 and shown in Fig. 6c, follows a different strategy. It uses a coarser 7 μm copper with almost no filler and a higher solids loading, and burns out for 2.8 h at 806 °C in 83% steam, some 40 K below the temperature at

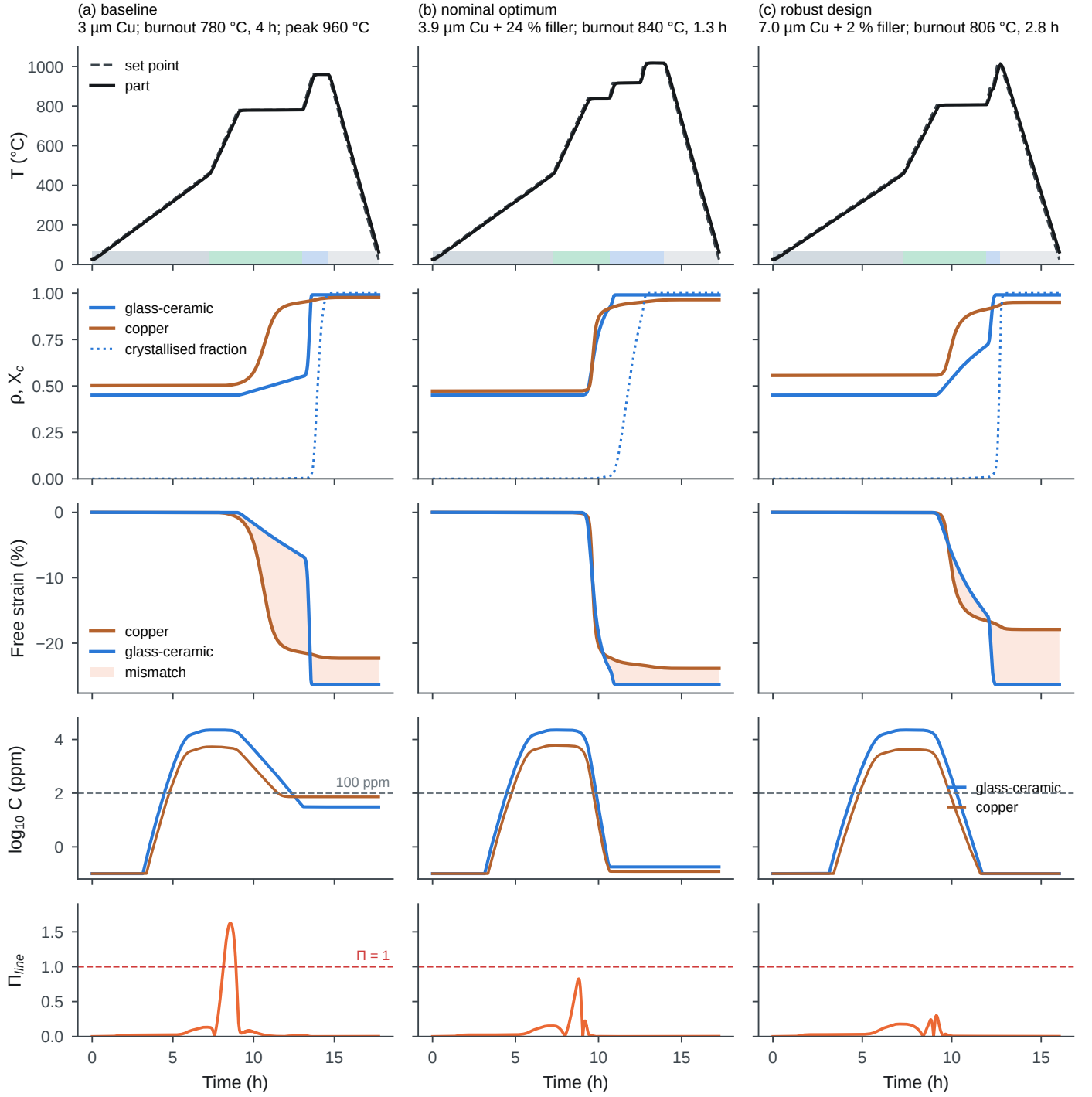


Figure 6: Co-firing histories of (a) the baseline, (b) the nominal optimum and (c) the robust design of Table 3. From top: furnace set point and part temperature, with the atmosphere of each segment shown by the band at the bottom (grey,  $\text{N}_2$ ; green, steam; blue, dry  $\text{H}_2/\text{N}_2$ ); relative densities and crystallised fraction of the glass; free logarithmic shrinkage of each material, the shaded area being the mismatch; carbon in the glass-ceramic and damage index of an embedded copper line.

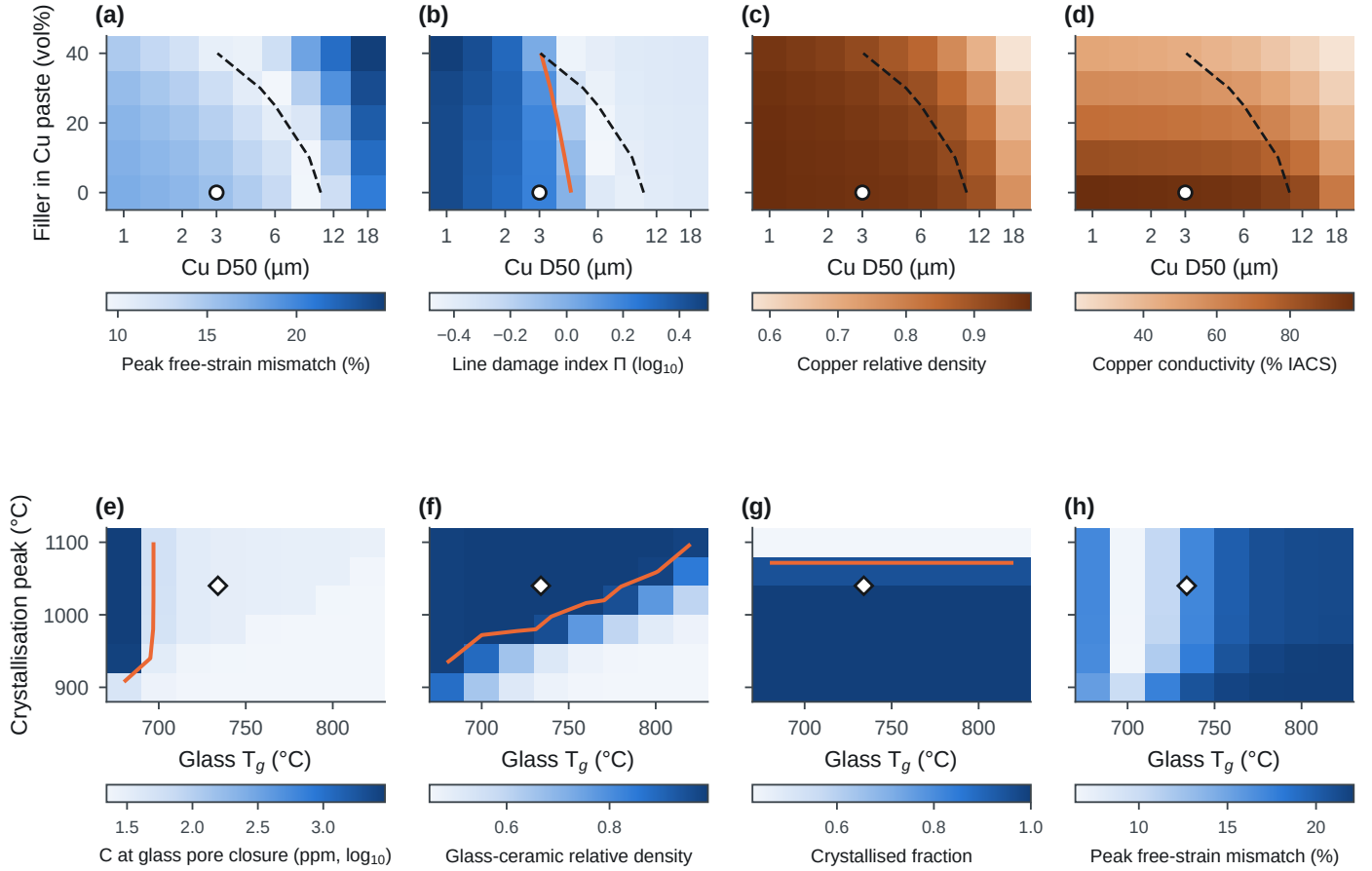


Figure 7: Design maps at the baseline programme. Top: copper paste, median particle size against filler fraction, with the baseline paste marked (circle): (a) peak free-strain mismatch; (b) damage index of an embedded line, orange contour  $\Pi = 1$ ; (c) copper density; (d) conductivity. The dashed line is the density limit  $\rho_{Cu} = 0.92$ ; pastes to its right fail it. Bottom: glass, transition temperature against crystallisation peak, with the calibrated glass marked (diamond) and the baseline paste: (e) carbon at pore closure, orange contour 100 ppm; (f) density of the glass-ceramic, orange contour 0.97; (g) crystallised fraction, orange contour 0.8; (h) peak mismatch.

which the glass would seal during the hold. It then heats quickly to a short peak at 1017 °C and needs 16.0 h in all. Both materials start to densify during the burnout hold, the copper once its char has gone and the glass because it is already soft. Being coarse, the copper runs only a few per cent ahead of the glass; the glass overtakes it during the heating to the peak. At nominal inputs the peak mismatch is 9.3%, the conductivity 89% IACS and the damage index 0.30. In both corners the carbon stays below 50 ppm, the damage index below 0.43 and the peak mismatch at or below 11.4%. The single-factor changes of Table 3 move its peak mismatch only between 9.3 and 9.9%. Robustness has a price: the nominal mismatch is two and a half times that of the sharp optimum, and the final camber,  $-187 \text{ m}^{-1}$ , is larger. Most of the remaining mismatch is a difference in final shrinkage, 8.4%, because the paste now carries more solids than the glass slurry (56 against 45 vol%), which helps the copper reach density without filler. A gasification rate four times slower than assumed defeats both designs, which leave 940 and 1295 ppm of carbon in the glass.

#### 4.8. The printed test vehicle

The three designs were then fired in three dimensions as the printed test vehicle of Fig. 1a: a  $12 \times 12 \times 2.4 \text{ mm}$  glass-ceramic block with a buried copper ground plane, a square spiral and a via to a top pad, meshed with 12 800 voxels of 0.3 mm and resting on a setter. Each run starts after binder removal and ends when the glass-ceramic is 98% dense, at 903–917 °C (Fig. 8). The setter keeps the lower face flat. The distortion therefore appears on the upper face, which dishes over the buried plane.

The buried plane dominates the result. In the baseline the copper has already densified when the glass begins to flow, so the plane acts as a rigid inclusion. The block shrinks by only 14.9% in plane, against 23.1% for the free glass-ceramic, and the glass above the plane, unable to shrink laterally, densifies by shrinking more in thickness. The upper face ends 622  $\mu\text{m}$  lower at the centre than at the rim (Fig. 8c), about a third of the fired thickness. In the nominal optimum the copper and the glass densify together. The plane then offers little resistance, the in-plane shrinkage rises to 21.2%, and the relief of the upper face falls to 150  $\mu\text{m}$ . The robust design lies between them: 18.3% in-plane shrinkage and 317  $\mu\text{m}$  of relief, consistent with its larger final mismatch. The via and pad raise a small plateau

Table 3: Baseline and optimised designs. Robustness: the nominal optimum and the robust design re-evaluated with the char gasification half-life ( $t_{1/2}$ , nominal 2 h) or the glass transition ( $T_g$ , nominal 734 °C) changed as indicated. Values that violate a constraint are in bold.

|                                                                   | Baseline       | Nominal optimum    | Robust design     |
|-------------------------------------------------------------------|----------------|--------------------|-------------------|
| <i>Paste and programme</i>                                        |                |                    |                   |
| Copper $d_{50}$ ( $\mu\text{m}$ ) / filler (vol%) / solids (vol%) | 3.0 / 0 / 50   | 3.9 / 24 / 47      | 7.0 / 2 / 56      |
| Burnout: $T$ (°C) / time (h) / steam (%)                          | 780 / 4.0 / 30 | 840 / 1.3 / 81     | 806 / 2.8 / 83    |
| Matching hold: $T$ (°C) / time (h)                                | –              | 916 / 1.7          | 889 / 0.2         |
| Peak: $T$ (°C) / time (h)                                         | 960 / 1.0      | 1018 / 1.2         | 1017 / 0.2        |
| Cycle time (h)                                                    | 17.7           | 17.3               | 16.0              |
| <i>Outcome at nominal inputs</i>                                  |                |                    |                   |
| Carbon at glass pore closure (ppm)                                | 33             | 0.2                | 0.03              |
| Glass-ceramic density / crystallised fraction                     | 0.991 / 1.00   | 0.990 / 1.00       | 0.990 / 1.00      |
| Copper density / conductivity (% IACS)                            | 0.976 / 96     | 0.965 / 64         | 0.950 / 89        |
| Peak free-strain mismatch (%)                                     | 15.6           | 3.7                | 9.3               |
| Line damage index $\Pi_{\text{Cu}}$                               | <b>1.63</b>    | 0.83               | 0.30              |
| Final camber of reference bilayer ( $\text{m}^{-1}$ )             | –254           | –76                | –187              |
| <i>Robustness: carbon at closure (ppm) / peak mismatch (%)</i>    |                |                    |                   |
| $t_{1/2} = 4$ h                                                   | –              | 76 / 11.1          | 37 / 9.9          |
| $t_{1/2} = 8$ h                                                   | –              | <b>1295</b> / 17.9 | <b>940</b> / 14.4 |
| $T_g - 15$ K                                                      | –              | 84 / 9.9           | 0.06 / 9.4        |
| $T_g + 15$ K                                                      | –              | 0.06 / 9.5         | 0.03 / 9.8        |

at the centre of every part. The ranking of the three designs by relief follows the ranking by mismatch of the one-dimensional analysis, which supports the use of that analysis for design. The absolute values, however, depend on the area of copper. A plane that covers 56% of the part, as here, is the most demanding case, and a real layout would divide it or add dummy metal in the rim.

#### 4.9. Cooling from the stress-free temperature

Below about 690 °C, the strain point of the calibrated glass, mismatch stresses no longer relax, and the expansion mismatch between copper (16.5–22 ppm K<sup>–1</sup> over this range) and the glass-ceramic (3 ppm K<sup>–1</sup>) is carried elastically. Fig. 9 shows the consequences for a copper plane on 1 mm of glass-ceramic. If copper remained elastic, its mean stress would exceed 900 MPa at room temperature and a 20 mm stack with 100  $\mu\text{m}$  of copper would bow by 250  $\mu\text{m}$ . Annealed copper cannot carry that stress. It yields within the first 10 K of cooling and then follows its yield stress, which rises from about 25 MPa at 690 °C to 69 MPa at room temperature. The bow falls accordingly, to 13  $\mu\text{m}$  for 100  $\mu\text{m}$  of copper and to 6  $\mu\text{m}$  for 50  $\mu\text{m}$ , and the glass-ceramic stays in mild compression (–7 MPa for 100  $\mu\text{m}$  of copper). Plastic accommodation makes the thermal-expansion mismatch a secondary problem compared with the sintering mismatch, provided the copper is dense and ductile. The price is cyclic plasticity in the conductor, which matters for thermal cycling in service but is outside the scope of this study.

#### 4.10. Which inputs decide the outcome

Fig. 10 ranks the fourteen uncertain inputs by their elementary effects around the nominal optimum. Elementary effects are expressed per full range of each factor. One input dominates. Across its range of 0.5–8 h, the half-life of char gasification changes the peak mismatch by 16 percentage points on average and the carbon at glass closure by several orders of magnitude, from well below 1 ppm to above the 100 ppm limit. It also comes

second for copper density, because char that is not removed keeps the copper from sintering. The carbon trapped in the glass depends next on the fragility and transition temperature of the glass, an order of magnitude less strongly. The density of the glass-ceramic is governed by its own properties: the transition temperature, the particle size and the crystallisation kinetics. The density and conductivity of the copper depend on the viscosity factor  $f_\eta$  and on the filler. The peak mismatch, the quantity that the design sought to minimise, responds to almost every input after the gasification rate:  $f_\eta$ , the glass transition, the char yield, and the particle size and solids loading of the glass. For every output that matters,  $\sigma$  is of the same order as  $\mu^*$ . The effects are therefore strongly non-linear and interact, which is what a design based on the coincidence of two processes should be expected to show. The tolerances on the copper paste itself (particle size  $\pm 30\%$ , filler  $\pm 5$  vol%) matter mainly for the damage index of a buried line, together with  $f_\eta$  and the char yield. The Avrami exponent and the activation energy of crystallisation matter only for the density of the glass-ceramic, and the crystallised fraction (not shown) remained complete in every run.

## 5. Discussion

### 5.1. The order of events in co-firing

Taken together, the results fix an order of events that any firing programme for this material pair has to respect. Pyrolysis in nitrogen leaves about 2% of char, relative to the solids, in the glass-ceramic and about 0.5% in the copper. The char must then be gasified before the glass seals its pores. With the atmosphere held at a fixed fraction of the copper–oxide equilibrium, this requires hours at 770–800 °C and is the slowest step of the process. Removing the char has a second effect, which is easy to overlook: it releases the copper. Residual carbon retards the sintering of copper, so a fine powder that has been held back by its own char starts to densify as soon as the steam clears

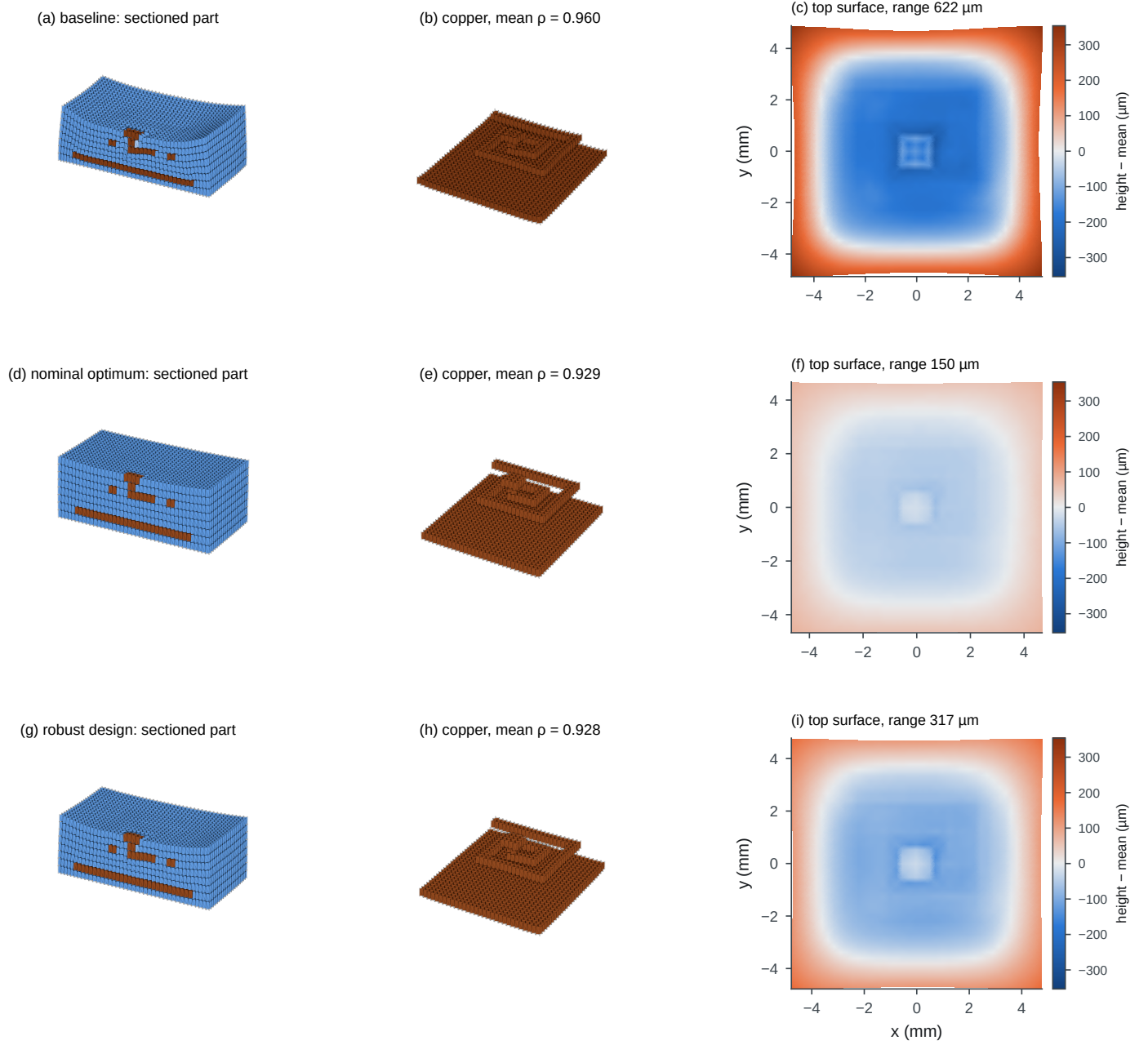


Figure 8: Three-dimensional co-sintering of the printed test vehicle (0.3 mm voxels, setter support, gravity) for the three designs of Table 3, from binder removal to 98% density of the glass-ceramic. Left: part sectioned through its centre, showing the buried plane, the spiral and the via. Centre: the copper, coloured by density. Right: height of the upper face relative to its mean; the lower face is held flat by the setter.

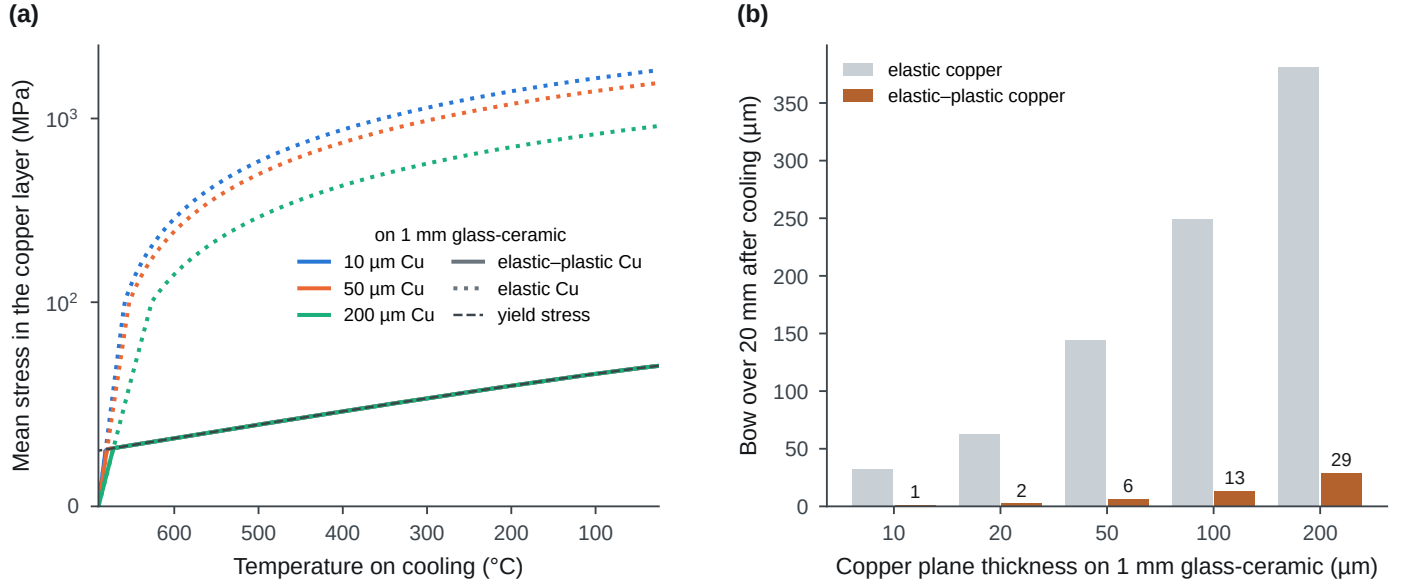


Figure 9: Cooling of a copper plane on 1 mm of glass-ceramic from the stress-free temperature (690 °C). (a) Mean stress in the copper layer for elastic and elastic-perfectly plastic copper; the plastic curves coincide with the temperature-dependent yield stress. (b) Bow over a 20 mm span at room temperature.

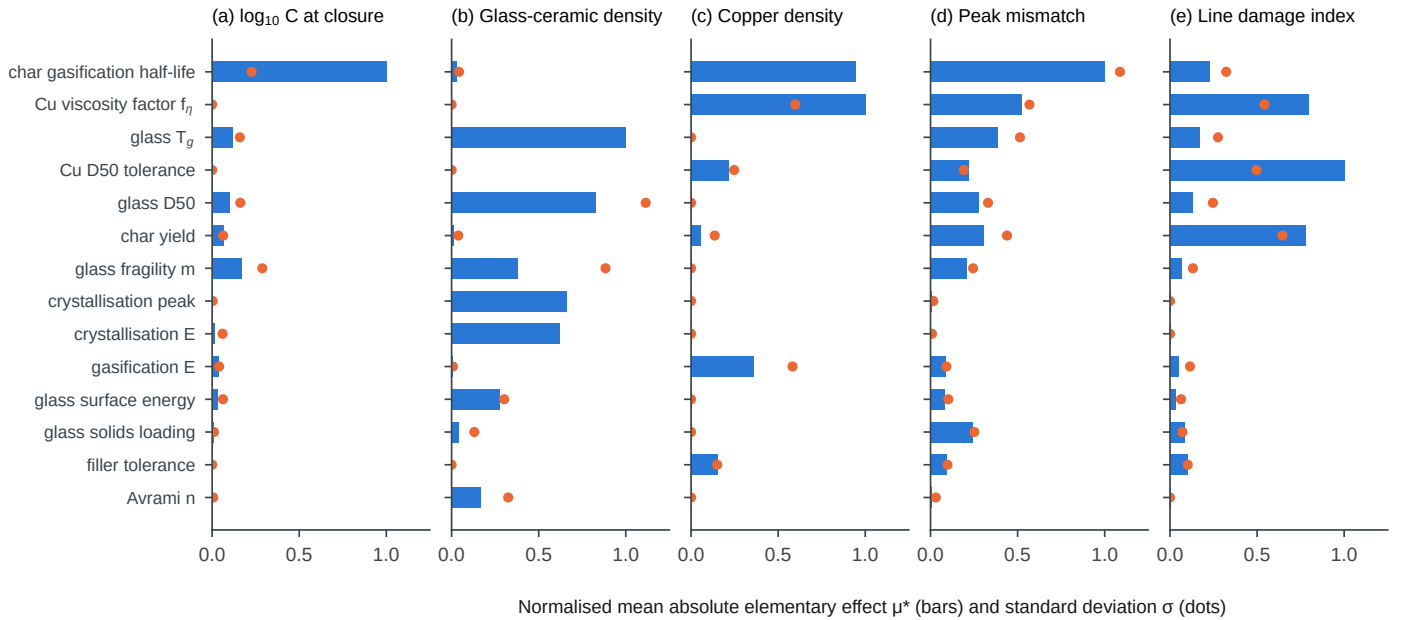


Figure 10: Morris screening of the fourteen uncertain inputs around the nominal optimum (16 trajectories, 240 co-firing evaluations). Bars: mean absolute elementary effect  $\mu^*$ ; dots: standard deviation  $\sigma$  of the elementary effects, both normalised by the largest  $\mu^*$  of each output. Factors are ordered by their summed normalised importance. Ranges as in Table 1; the copper particle size and filler fraction are varied as manufacturing tolerances around the design.

it, at 680–780 °C in the baseline and well before the glass. In silver-metallised LTCC fired in air this coupling does not arise, because the binder burns off completely at low temperature and the metal is free to sinter from then on. Here the burnout step is also, whether intended or not, the first sintering step of the conductor. The glass then densifies over a range of about 80 K and stops when it crystallises, and below the strain point of the residual glass the expansion mismatch is taken up by plastic flow of the copper.

This order explains why neither lever works alone. The furnace programme cannot delay the densification of fine copper beyond the burnout, because the burnout has to happen below the coalescence of the glass. The paste cannot be made as late and as steep as the glass (Fig. 5) without losing conductor density. The two designs of Table 3 are the two ways out. The nominal optimum uses the coupling between char and copper to start both materials together. It reaches the lowest mismatch but depends on the timing of char removal, which is the least known input of the model. The robust design gives up part of that gain. It relies on a coarse copper that is intrinsically slow, and it tolerates a factor of two in the gasification rate together with 15 K in the glass transition. Which of the two is preferable depends on what has been measured. Once the gasification kinetics and the viscosity of the glass are known to within these bounds, a design closer to the nominal optimum becomes safe.

## 5.2. Relation to tape-based co-firing

The rule used for the atmosphere, hydrogen tied to steam so that the water-to-hydrogen ratio sits at a fixed fraction of the copper limit, reproduces the operating point of the IBM process (Fig. 3), and the calibrated glass reproduces the reasons given for its burnout temperature (Table 2). The model does not add a new principle to that process. What it adds is a quantitative account: how long the burnout must last at a given temperature and steam content, how much carbon is left when the pores close, and how the choice of copper powder interacts with the programme. In silver-based LTCC, camber is controlled empirically by adjusting the paste [38] or the heating rate [39]. For copper in a crystallising glass the ladder of Fig. 5 shows why paste formulation alone is not sufficient. The heating programme sets the order in which the materials are released to sinter, and the paste only fine-tunes the rate once they are.

Printing changes the problem in three respects. First, a photocurable slurry carries more binder than a cast tape: at 45 vol% solids the glass-ceramic shrinks by 23% linearly, against 12.7–15.8% in the plane of commercial LTCC tapes [81, 82], and it holds correspondingly more char. Second, conductors are no longer confined to printed planes connected by punched vias. The test vehicle contains a spiral, a buried plane and a via defined in one build, and the three-dimensional model shows that their densification and distortion follow from the same constitutive law as in a bilayer. It also shows that a large buried plane, which tape technology would mesh or perforate, dominates the distortion; a printed layout can divide such a plane freely. Third, printed ceramics shrink anisotropically between the build direction and the plane, as laminated tapes do: unconstrained IBM laminates fired in H<sub>2</sub>/H<sub>2</sub>O shrink about half as much through the

thickness as in the plane [79]. The model treats both materials as isotropic. Anisotropy would change the camber and the shape of the fired part, but not the order of events described above.

## 5.3. Limitations

The results rest on published data and a model; no part has been fired. Three inputs deserve particular caution. The viscosity of the glass was calibrated against statements in patents, not measured, and it describes one glass composition; a different cordierite-type glass will shift the whole window. The gasification kinetics of photopolymer char on either material are unknown and were estimated, so the absolute times of the burnout carry the largest uncertainty of the study (Section 4.10). The viscosity factor of copper was fitted to a coarser powder than those considered here. In addition, the model omits several phenomena that can matter in practice: dissolution of copper oxide in the glass and interdiffusion at the interface, adhesion and delamination, the water released by the glass during crystallisation, the effect of residual carbon on dielectric loss, and cyclic plasticity of the copper in service. The damage indices compare stresses with strength estimates and are meant for ranking designs, not for predicting failure. The three-dimensional runs stop at 98% density of the glass-ceramic, so the final densification of the copper at the peak temperature, inside an already rigid body, is represented only in the slab model.

## 5.4. What to measure first

The sensitivity analysis (Fig. 10) turns the limitations into a short list of measurements, ordered by their value to the design:

1. The gasification kinetics of the photopolymer char on each material, in the steam–hydrogen atmosphere of the burnout. Interrupted burnout runs followed by carbon analysis at three temperatures would fix both the rate and its activation energy. No other quantity has as large an effect on the carbon trapped in the glass and on the mismatch.
2. The viscosity of the glass between 750 and 900 °C, which fixes  $T_g$  and  $m$ . Beam bending or parallel-plate viscometry of the glass would give it directly, and so would isothermal dilatometry of the printed glass-ceramic, interpreted with the closed form of Appendix B.
3. The sintering viscosity of the copper slurry,  $f_\eta$ , from dilatometry of printed copper coupons in dry hydrogen after binder removal.
4. The char yield of both cured slurries, from thermogravimetry in nitrogen.

The particle size and solids loading of the glass also matter, but they are specified by the supplier and the slurry formulation. The crystallisation kinetics matter little within the ranges considered, as long as the exotherm peak lies above the sintering range. All of these measurements are routine, and most of them can be made on single-material coupons printed with the same slurries.

## 5.5. Implications for three-dimensional packages

A fired body of this kind has the properties that made glass-ceramic/copper attractive for high-end packaging: a thermal expansion of 3.0 ppm K<sup>-1</sup>, close to that of silicon (2.6 ppm K<sup>-1</sup>)

[71], a permittivity of about 5 [62], and copper conductors. The model predicts 96% IACS for the conductor from its porosity alone. IBM reported  $2.0 \pm 0.2 \mu\Omega \text{ cm}$ , about 86% IACS, for copper co-fired in  $\text{H}_2/\text{H}_2\text{O}$  [62]; the difference points to impurities and grain-boundary oxygen, which the model does not include. Printing adds the geometric freedom that tape lamination lacks. Vertical coils, shielded lines, cavities and ground planes of arbitrary shape can be built into one part, which is relevant to antenna-in-package modules and millimetre-wave front ends [3, 2] and to substrates for wide-bandgap power devices [4]. The present study defines a candidate firing route for such parts and the few quantities that have to be measured before it is tried.

## 6. Conclusions

A model of binder removal, sintering and crystallisation was used to design the co-firing of copper inside a cordierite-type glass-ceramic printed as one part by multi-material vat photopolymerisation. It was verified against closed-form solutions, and its glass was calibrated against the published IBM glass-ceramic/copper process, whose atmosphere and burnout temperature it reproduces. No experiments were made. The main conclusions are as follows.

1. Of the atmospheres examined, only steam removes the carbon from the glass before its pores close. Hydrogen tied to the steam at a fixed fraction of the  $\text{Cu}/\text{Cu}_2\text{O}$  equilibrium keeps the copper metallic and reproduces the IBM operating point. At  $780^\circ\text{C}$  such an atmosphere gasifies char seven times faster than humidified forming gas, which leaves more than 300 ppm of carbon in the glass-ceramic in every programme examined.
2. Fine copper and the glass densify in different temperature ranges:  $450\text{--}700^\circ\text{C}$  for  $3 \mu\text{m}$  copper against  $850\text{--}930^\circ\text{C}$  for the glass. Co-fired with the IBM-type programme, the copper densifies during the burnout, the free-strain mismatch reaches 15.6%, and a buried line is overloaded in tension. The final shrinkages differ by only 4%; the difficulty lies in the order in which the two materials shrink.
3. Char holds copper back, so the burnout step also releases the copper to sinter. An optimised paste and programme exploit this coupling: a short, hot burnout just below the coalescence of the glass releases both materials together and reduces the peak mismatch to 3.7%. The coincidence is sharp. A factor of two in the gasification rate, or 15 K in the glass transition, restores most of the mismatch, and the 24% filler it requires limits the copper to 64% IACS.
4. A design required to meet every constraint for a factor of two either way in the gasification rate, combined with  $\pm 15 \text{ K}$  in the glass transition, uses  $7 \mu\text{m}$  copper with almost no filler and a 2.8 h burnout at  $806^\circ\text{C}$ . Across that uncertainty it keeps the peak mismatch between 9.3 and 11.4%, the copper at 89% IACS and the carbon at glass closure below 50 ppm, in a 16 h cycle. Robustness costs a factor of two and a half in nominal mismatch.
5. In three dimensions, a buried copper plane that densifies before the glass acts as a rigid inclusion. The baseline test vehicle shrinks by 15% in plane instead of 23%, and its upper

face dishes by 0.6 mm. The nominal optimum reduces the dishing to 0.15 mm and the robust design to 0.32 mm, the same ranking as the one-dimensional mismatch.

6. On cooling, plastic flow of the copper limits the stress it transmits to its yield stress. A  $100 \mu\text{m}$  copper plane on 1 mm of glass-ceramic bows by  $13 \mu\text{m}$  over 20 mm, about a nineteenth of the elastic prediction.
7. The inputs that decide the outcome are, in order, the gasification kinetics of the char, the viscosity of the glass, the sintering viscosity of the copper and the char yield. All four can be measured on single-material printed coupons, and they should be measured before the process is tried.

## CRediT authorship contribution statement

**[Author names]:** Conceptualization, Methodology, Software, Formal analysis, Visualization, Writing – original draft, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The simulation code, input parameters with their provenance, and the scripts that generate every figure in this paper are available from the corresponding author on reasonable request.

## Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used [name of tool] in order to [reason, e.g. assist with code development, data analysis and drafting of the text]. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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*Crystallisation kinetics.* With the rate constant calibrated as described in Section 2.3, the simulated exotherm at  $10 \text{ K min}^{-1}$  peaks within 0.01 K of the prescribed temperature for Avrami exponents of 1, 2 and 3.

## Appendix A. Copper slab model and its calibration

The copper conductor uses the slab model developed for single-material copper parts printed from a photocurable slurry. Besides the binder chemistry of Section 2.2, it tracks the oxygen bound in the powder as  $\text{Cu}_2\text{O}$ . The oxide grows in oxygen-bearing gas by a decelerating law fitted to powder oxidation data, is reduced by hydrogen above the equilibrium ratio of Eq. (9), and reacts with char by  $\text{Cu}_2\text{O} + \text{C} \rightarrow 2 \text{Cu} + \text{CO}$ . None of these oxidation paths is active in the co-firing programmes studied here, because the inlet gas contains no oxygen and the water-to-hydrogen ratio is held below the copper–oxide equilibrium. What matters for co-firing is the sintering viscosity. Eq. (7) uses the lattice and grain-boundary diffusion data for copper tabulated by Frost and Ashby [52], with one factor,  $f_\eta$ , fitted to the densification of copper parts printed from a 60 vol% photocurable slurry of  $22 \mu\text{m}$  powder and sintered for 4 h at  $1050^\circ\text{C}$  in hydrogen, which reached 90–94% relative density [31]. The fit gives  $f_\eta = 0.28$ , with 0.22–0.34 spanning the reported density range. The powder size enters Eq. (7) through the grain size, set initially to 0.4 times the median particle size, and through the sintering stress. Transferring  $f_\eta$  from  $22 \mu\text{m}$  to the 3–12  $\mu\text{m}$  powders considered here therefore rests on the diffusional scaling of Eq. (7); this is why  $f_\eta$  is included in the sensitivity analysis of Section 4.10.

## Appendix B. Verification

*Viscous densification of the glass-ceramic.* With crystallisation and char removed, and open porosity, Eq. (3) with a Newtonian matrix gives  $\dot{\theta} = -9\gamma\theta/(4r\eta)$ . The slab model was run through isothermal holds at 810, 830 and  $850^\circ\text{C}$  with 4, 8 and 16 cells. The simulated porosity follows the closed form, integrated along the simulated part temperature, to within  $1.3 \times 10^{-4}$  over the full range of densification before pore closure, independently of the mesh (Fig. 2a).

*Bilayer curvature.* The three-dimensional model of a free  $12 \times 12 \text{ mm}$  plate made of 0.6 mm copper on 1.8 mm glass-ceramic was solved for the instantaneous velocity field at  $900^\circ\text{C}$ , and the curvature rate was fitted to the mid-plane deflection. It converges to the viscous Timoshenko value of Eq. (10) with second-order behaviour: the error falls from 3.4% at 0.6 mm voxels to 0.69, 0.22 and 0.05% at 0.3, 0.2 and 0.15 mm (Fig. 2b).

*Free sintering and mass conservation.* A free, frictionless cube held isothermally densifies homogeneously and isotropically, matches the closed-form Skorohod–Olevsky solution to 0.22% in density, and conserves solid mass to 0.07%.